Chudi Zhong

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ACADEMIC POSITION

University of North Carolina at Chapel Hill — Chapel Hill, NC
Assistant Professor — 07/2024 – Present
School of Data Science and Society
Department of Statistics and Operations Research

EDUCATION

Duke University — Durham, NC
Ph.D. in Computer Science — 08/2020 – 05/2024
Thesis: Interpretability and Multiplicity: a Path to Trustworthy Machine Learning

M.S. in Statistical Science — 08/2018 – 05/2020

University of North Carolina at Chapel Hill — Chapel Hill, NC
B.S. in Statistics and Information Science — 08/2014 – 05/2018

RESEARCH OVERVIEW

My research focuses on (1) developing **interpretable machine learning** algorithms and pipelines to facilitate human-model interaction for high-stakes decision-making problems and (2) learning the **Rashomon Effect**, the existence of many distinct yet almost equally accurate models, and its impact on modern large models.

AWARDS

Second Place, Bell Labs Prize, 2023
The Bell Labs Prize is an international competition to solicit game changing and impactful ideas that have the potential to change the way people live, work, and communicate with each other. Among 107 proposals received in 2023, our team won the second place in the Bell Labs Prize.
Rising Stars in Data Science, University of Chicago, 2023
Outstanding Ph.D. Preliminary Research Award, Duke Computer Science, 2023
Outstanding Research in Progress-RIP Award, Duke Computer Science, 2022
Finalist, Data Mining Best Student Paper Award, INFORMS, 2022

PUBLICATIONS

† students, * equal contribution

PUBLISHED

[13] Ethan Hsu†, Harry Chen†, **Chudi Zhong**, Lesia Semenova. The Double-Edged Nature of the Rashomon Set for Trustworthy Machine Learning. In *International Conference on Machine Learning (**ICML**)*, 2026. (**Spotlight**, top 8.4% in all accepted papers, top 2.2% in all submissions)

[12] Ethan Hsu†, Tony Cao†, Lesia Semenova, **Chudi Zhong**. The Rashomon Set Has It All: Analyzing Trustworthiness of Trees under Multiplicity. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2025.

[11] **Chudi Zhong**, Panyu Chen, Cynthia Rudin. Models that are Interpretable but not Transparent. In *International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, 2025.

[10] Cynthia Rudin, **Chudi Zhong**, Lesia Semenova, Margo Seltzer, Ronald Parr, Jiachang Liu, Srikar Katta, Jon Donnelly, Harry Chen, Zachery Boner. The Amazing Things that Come from Having Many Good Models. In *International Conference on Machine Learning (**ICML**)*, 2024. (**Spotlight**, top 12.8% in all accepted papers, top 3.5% in all submissions)

[9] **Chudi Zhong***, Zhi Chen*, Jiachang Liu, Margo Seltzer, Cynthia Rudin. Exploring and Interacting with the Set of Good Sparse Generalized Additive Models. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2023.

[8] Jiachang Liu, Sam Rosen, **Chudi Zhong**, Cynthia Rudin. OKRidge: Scalable Optimal k-Sparse Ridge Regression for Learning Dynamical Systems. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2023. (**Spotlight**)

[7] Rui Xin*, **Chudi Zhong***, Zhi Chen*, Takuya Takagi, Margo Seltzer, Cynthia Rudin. Exploring the Whole Rashomon Set of Sparse Decision Trees. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022. (**Oral**, top 6.3% in all accepted papers, top 2% in all submissions)
- **Finalist, Data Mining Best Student Paper Award, INFORMS, 2022**

[6] Jiachang Liu*, **Chudi Zhong***, Boxuan Li, Margo Seltzer, Cynthia Rudin. FasterRisk: Fast and Accurate Interpretable Risk Scores. In *Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022.

[5] Zijie Wang, **Chudi Zhong**, Rui Xin, Takuya Takagi, Zhi Chen, Duen Horng Chau, Cynthia Rudin, Margo Seltzer. TimberTrek: Exploring and Curating Trustworthy Decision Trees with Interactive Visualization. In *IEEE Visualization and Visual Analytics (**VIS**)*, 2022.

[4] Jiachang Liu, **Chudi Zhong**, Margo Seltzer, Cynthia Rudin. Fast Sparse Classification for Generalized Linear and Additive Models. In *International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, 2022.

[3] Hayden McTavish*, **Chudi Zhong***, Reto Achermann, Ilias Karimalis, Jacques Chen, Cynthia Rudin, Margo Seltzer. Fast Sparse Decision Tree Optimization via Reference Ensembles. In *AAAI Conference on Artificial Intelligence (**AAAI**)*, 2022.

[2] Cynthia Rudin, Chaofan Chen, Zhi Chen, Haiyang Huang, Lesia Semenova, **Chudi Zhong**. Interpretable Machine Learning: Fundamental Principals and 10 Grand Challenges. *Statistics Surveys*, 2022.

[1] Jimmy Lin*, **Chudi Zhong***, Diane Hu, Cynthia Rudin, Margo Seltzer. Generalized and Scalable Optimal Sparse Decision Trees. In *International Conference on Machine Learning (**ICML**)*, 2020.

PREPRINT

1 Shihan Feng†, Cheng Zhang†, Michael Xi†, Ethan Hsu†, Lesia Semenova, **Chudi Zhong**. Many Ways to be Right: Exploring the Rashomon Set for Concept-Based Models. Under Review.

2 Yuexin Zhang†, Joseph Lavond†, **Chudi Zhong**, Yao Li. Partition-Losses Fine-Tuning: Contamination-Robust Backdoor Unlearning. Under Review.

3 Shihan Feng†, Xin Zheng†, Can Chen, Ren Wang, **Chudi Zhong**. Interpretable Hypergraph Learning. Under Review.

GRANTS

UNC School of Data Science and Society Research Clusters Seed Grant: Uncovering Gene Pathways through Multimodal Hypergraph Contrastive Learning. Awarded. Total amount: $15,000. Core faculty: Di Wu, Can Chen, Weitong Zhang, **Chudi Zhong**, Qingyun Liu. 2025-2026.

TEACHING

DATA 521: Foundations in AI	Fall 2026
STOR 565: Machine Learning	Fall 2025
DATA 110: Introduction to Data Science	Spring 2025
STOR 520: Statistical Computing for Data Science	Fall 2024

MENTORING

PhD STUDENTS

Lewis Dubrowski (UNC STOR; co-advised with Nilay Argon, Serhan Ziya)	2024-
Cheng Zhang (UNC CS)	2025-

MS STUDENTS

Isabel Giovannucci (UNC; co-advised with Ali Parlakturk)	2026-
Yinyu Yao (UNC; co-advised with Ali Parlakturk)	2025-
Yuexin Zhang (UNC; co-advised with Yao Li; now PhD student at University of Pittsburgh)	2024-2025
Shihan Feng (Duke; incoming PhD student at UNC Chapel Hill)	2024-2026
Xin Zheng (Duke; incoming PhD student at UNC Chapel Hill)	2024-2026
Panyu Chen (Duke; now PhD student at University of College London)	2024-2025

UNDERGRADUATES

Prithika Roy (UNC; co-advised with Serhan Ziya; incoming PhD student at UC Berkeley)	2024-
Tommy Smith (UNC; co-advised with Nilay Argon, Serhan Ziya, incoming PhD student at UNC Chapel Hill)	2024-
Harish Pandyaram (UNC)	2024-2025
Bo Chi (Duke; co-advised with Cynthia Rudin)	2026-
Frank Zhang (Duke; co-advised with Cynthia Rudin)	2026-
Chris Li (Duke; co-advised with Cynthia Rudin)	2025-
Ethan Hsu (Duke; graduate with Alex Vasilos award; now working in industry)	2024-2026
Tony Cao (Duke)	2024-2025
Rui Xin (Duke; now PhD student at University of Washington)	2021-2023
Boxuan Li (Duke; now PhD student at Columbia University)	2021-2023
Diane Hu (Duke; now working in industry)	2019-2020

THESIS/PRELIM COMMITTEES (not including my students)

Thesis committee for UNC STOR masters student Sean Shen	2026
Thesis committee for UNC STOR masters student Akshata Rai	2026
Thesis committee for UNC STOR masters student Varshith Tipirneni	2026
Thesis Reader for UNC CS masters student Pavan Akkineni	2026
Prelim committee for UNC STOR PhD student Dilay Ozkan	2025
Prelim committee for UNC STOR PhD student Kyung Rok Kim	2025
Prelim committee for UNC STOR PhD student Hyeon Lee	2025

INVITED TALKS

Statistics and Data Science Seminar, National University of Singapore, Singapore	2026
Economics Seminar, Nangyang Technological University, Singapore	2026
INFORMS Annual Meeting	2025
Data Science Day Short Talk, UNC-Chapel Hill	2025
Computational Medicine Series, UNC-Chapel Hill	2025
FRG Series, UNC-Chapel Hill	2024
School of Business, Purdue University	2024
School of Information Sciences Colloquium, University of Illinois-Urbana Champaign	2024
ASSET Seminar, University of Pennsylvania	2024
Statistics and Operations Research Colloquium, UNC-Chapel Hill	2024
Computer Science and Philosophy AI Workshop, UNC-Chapel Hill	2023
Rising Stars in Data Science, University of Chicago	2023
Tippie College of Business, University of Iowa	2023
INFORMS Annual Meeting	2023
Cornell ORIE Young Researcher Workshop (poster), Cornell University	2023
Publisher-centric Science Talks, Amazon	2023
Interpretability in AI Workshop, Banff International Research Station	2022
INFORMS Annual Meeting	2021

EXTERNAL SERVICE

Workshop Organizer:
- "Navigating the Model Uncertainty and the Rashomon Effect: From Theory and Tools to Applications and Impact", AAAI Conference on Artificial Intelligence (AAAI), January, 2026, Singapore.

Tutorial Organizer:
- "The Illusion of the Best Model – Multiplicity, Interpretability, and Accountability in High-Stakes AI". ACM Conference on Fairness, Accountability, and Transparency (FAccT), June, 2026.
- "From Underspecification to Alignment: Breaking the One-Model Mindset for Reliable AI". AAAI Conference on Artificial Intelligence (AAAI), January, 2026.
- "Making Models We Can Understand: An Interactive Introduction to Interpretable Machine Learning". International Conference on Computational Social Science (IC2S2), July, 2024.

Panelist:
- Harnessing AI for Material Symposium, Duke University, 2024

Session Chair:
- "Advancing Interpretable Machine Learning: Novel Approaches and Applications", INFORMS Annual Meeting, 2023, Phoenix, USA.

Conference Area Chair:
- Interpretable AI Workshop, Advances in Neural Information Processing Systems (NeurIPS), 2024.

Conference Reviewer:
- ICML (2026, 2025, 2022)
- NeurIPS (2024, 2023)
- AAAI (2025)
- AISTATS (2023, 2022, 2021)
- CVPR (2026)
- ECCV (2026)
- COLT (2023)

Journal Reviewer:
- Annals of Applied Statistics
- Journal of the Royal Statistical Society Series B
- Operations Research
- Journal of Computational and Graphical Statistics
- Journal of Machine Learning Research

SERVICE TO UNC

SDSS undergraduate advising, 2024-2026
STOR Departmental Retreat Planning Committee, 2024-2026
STOR Ethical AI Faculty Search Committee, 2025
Development of a new advanced undergraduate level course, DATA 521: Foundations in AI.
Development of the AI concentration and data engineer track for BS data science.
Serve on the SDSS Data + AI Service Faculty Advisory Group.